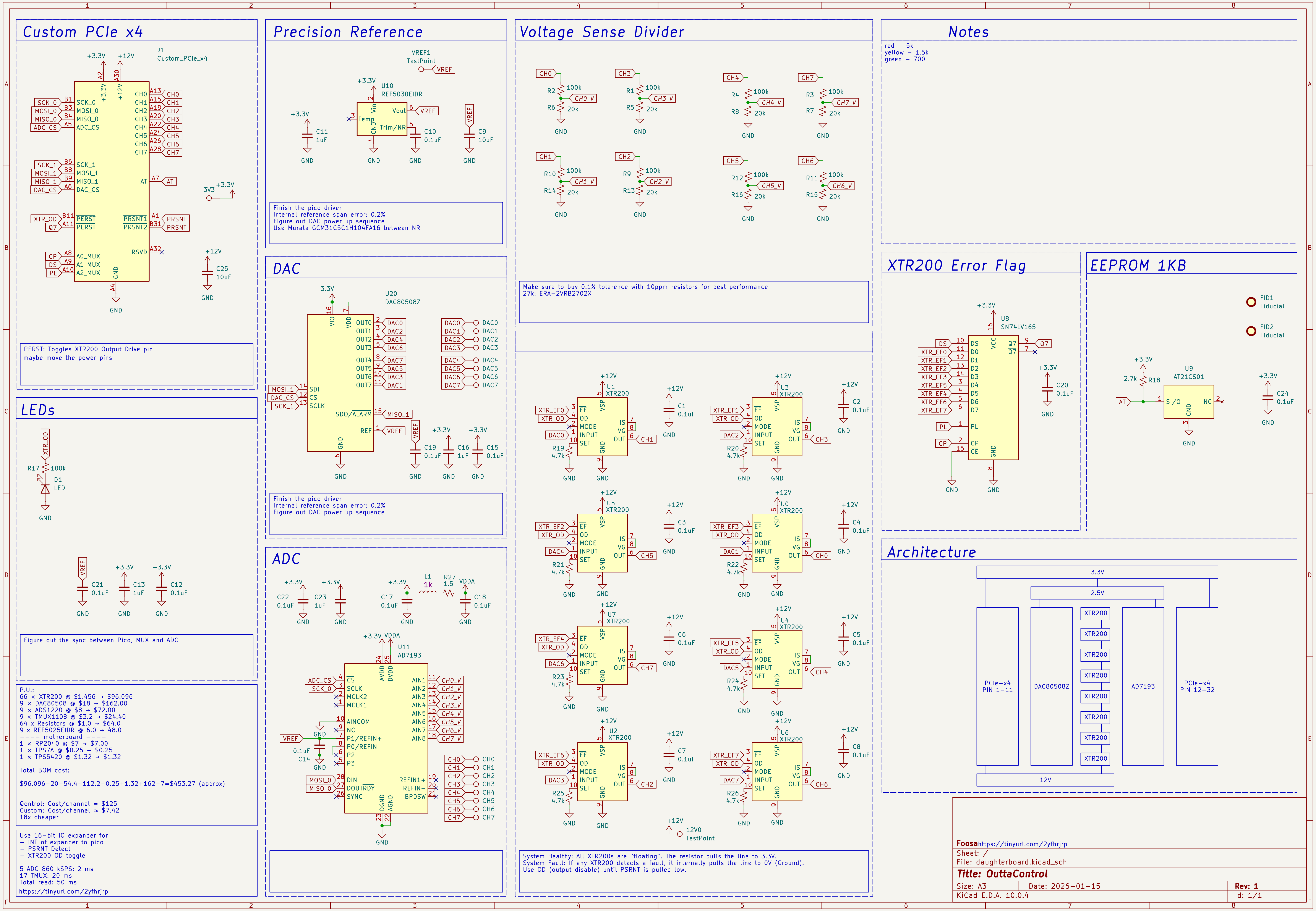
Custom PCIe x4
J1
Custom_PCIe_x4
PERST: Toggles XTR200 Output Drive pin
maybe move the power pins
Precision Reference
VREF1
TestPoint
U10
REF5030EIDR
Finish the pico driver
Internal reference span error: 0.2%
Figure out DAC power up sequence
Use Murata GCM31C5C1H104FA16 between NR
Voltage Sense Divider
Make sure to buy 0.1% tolarence with 10ppm resistors for best performance
27k: ERA-2VRB2702X
Notes
red - 5k
yellow - 1.5k
green - 700
DAC
U20
DAC80508Z
Finish the pico driver
Internal reference span error: 0.2%
Figure out DAC power up sequence
LEDs
Figure out the sync between Pico, MUX and ADC
ADC
U11
AD7193
XTR200 Error Flag
U8
SN74LV165
EEPROM 1KB
U9
AT21CS01
FID1
Fiducial
FID2
Fiducial
Architecture
3.3V
2.5V
XTR200
PCIe-x4 PIN 1-11
DAC80508Z
AD7193
PCIe-x4 PIN 12-32
12V
12V0
TestPoint
System Healthy: All XTR200s are "floating". The resistor pulls the line to 3.3V.
System Fault: If any XTR200 detects a fault, it internally pulls the line to 0V (Ground).
Use OD (output disable) until PSRNT is pulled low.
P.U.:
66 × XTR200 @ $1.456 → $96.096
9 × DAC80508 @ $18 → $162.00
9 × ADS1220 @ $8 → $72.00
9 × TMUX1108 @ $3.2 → $24.40
64 x Resistors @ $1.0 → $64.0
9 x REF5025EIDR @ 6.0 → 48.0
---- motherboard ----
1 × RP2040 @ $7 → $7.00
1 × TPS7A @ $0.25 → $0.25
1 × TPS5420 @ $1.32 → $1.32
Total BOM cost:
$96.096+20+54.4+112.2+0.25+1.32+162+7=$453.27 (approx)
Qontrol: Cost/channel = $125
Custom: Cost/channel ≈ $7.42
18x cheaper
Use 16-bit IO expander for
- INT of expander to pico
- PSRNT Detect
- XTR200 OD toggle
5 ADC 860 kSPS: 2 ms
17 TMUX: 20 ms
Total read: 50 ms
https://tinyurl.com/2yfhrjrp
Foosa https://tinyurl.com/2yfhrjrp
Sheet: /
File: daughterboard.kicad_sch
Title: OuttaControl
Size: A3
Date: 2026-01-15
Rev: 1
KiCad E.D.A. 10.0.4
Id: 1/1